

Build a strong foundation and anything you build on top will be strong too.
Tested here are 48 motherboards and 16 processors to help you select the right materials
for your system's foundation

Inquire about the computer system that a friend or colleague has and the answer would likely revolve around the processor within. It is not surprising to find out that the majority of users make their buying decisions around processors. Everything else comes later. But what they are forgetting is the motherboard, the platform which contributes to a major part of the overall performance. Like a solid foundation to a house and a good chassis to an engine, a motherboard is vitally important to the whole picture. It can enhance, or hold back, the performance capability of a processor. Coupling an extremely fast processor with a mediocre board could seriously hamper performance just as using a low-end processor on a feature-rich, high-end board would.

Another major concern is the main area of application that you would be working with on your system. Do you really need a Rs 38,600 Intel Pentium 4 3.06 GHz processor to run Microsoft Excel? The theory of overkill would conclude the answer to be a no. Potential buyers should, therefore, not make the mistake of buying components without thinking about the applications they'll be using. We hope that this article will help you make the right choice whether you are in the market for a processor, a motherboard or for a complete solution. To do so, we present a comprehensive test that covers 48 motherboards and 16 processors from the stables of the major players, Intel and AMD.

Market analysis

The Indian PC market is always buzzing with activity. The reason for this is the easy availability of a wide range of components—in this case, processors and motherboards. The ever-decreasing prices due to stiff competition have led many users to upgrade to comparatively powerful CPUs such as the P4 or the

Athlon XP range of processors. The same goes for motherboards too—you will be amazed by the sheer choice you have now, with boards based on different types of chipsets available for different processors.

Going by the current market trend, what we can conclude is that with the demise of the P-III processors for Intel and the Duron processors for AMD, users have now shifted their focus to buying processors and motherboards that hit the sweet spot in price and can be pushed for maximum performance gain. One example of this is the marriage of the P4 1.6 GHz (Northwood core) processor with the various boards such as the ASUS P4B266 and MSI 845PE Max 2 to name a few. This combination created waves in the enthusiast user community, because the P4 1.6 GHz could be easily pushed to 2.4 GHz without compromising the system's stability. Hence, though the boards were competitively priced, they also offered immense tweaking capabilities—an irresistible combination for the value-minded buyer.

How we tested

To test a processor's performance, we evaluated it using tools that gave us both a theoretical and real world picture. The test machine consisted of an Intel D845PEBT2 for Intel-based processors and the MSI KT4 Ultra for AMD chips. We used 256 MB of PC2700 333MHz DDR memory from Corsair. The ATI Radeon 9700 Pro was our choice of graphics card for the system and the Seagate Barracuda ATA IV 40 GB was used as the test hard disk drive. The monitor was a Samsung 75E 17-inch and the OS used was Windows XP Professional. For testing the performance of the processors, we used different CPU-intensive tests such as 3DMark 2001SE, Specview Perf (a 3D modelling benchmark) and POV-Ray, which is an extremely CPU-

STTI Scoring

Based on its performance, and married with its price, each processor and motherboard was awarded an STTI (Suitability To Task Index) score. We chalked up three application areas and ranked the components tested (based on the STTI score) within each of these sectors: Gaming and Entertainment (performance 90 per cent, price 10 per cent weightage), Content Creation (performance 75 per cent, price 25 per cent weightage) and Office and Internet (performance 50 per cent, price 50 per cent weightage).

The area of Gaming and Entertainment is for those of you who wish to watch high-quality DVD movies or play next-generation computer-intensive games. Budget is not a major consideration within this category and we awarded just a 10 per cent weightage to price within this sector. Under Content Creation fall tasks such as working with Adobe Photoshop 7, 3ds max5, Maya, Adobe Premiere, etc. Price is of a greater importance here and is awarded a weightage of 25 per cent, performance getting the remainder 75 per cent. Office and Internet is the final category to consider. E-mail, chatting, Net browsing and document creation under Microsoft Office or other office suites, are some of the basic tasks that fall under this area. It is thus a very budget-conscious section

where cost shares a hefty 50 per cent weightage with performance.

The STTI index provided a clear indication of the performance of the processors and motherboards in these application areas. For example, the STTI for Content Creation was calculated by allotting a greater weightage to tests such as the Photoshop 7.0 filter test than to the gaming tests. Similarly, for Gaming and Entertainment, a greater importance was allotted to the performance of a CPU under *Quake III Arena* and 3DMark 2001 SE; or the presence of features such as onboard RAID and six-channel integrated audio on a motherboard. Before making a buying decision, the first thing you need to define is the most common tasks you wish to perform using your computer. Then choose the relevant application area and within the test analysis, you will find the best processors and the best motherboards for your job.

Any processor or motherboard that distinguished itself by scoring an STTI of more than 70 under each of the above areas should be considered well worth buying. Finally, to present a comprehensive solution, we paired up the best processor and motherboard from within each application area (the best were the highest scorers, of course)—this pair would then be the best solution for the task at hand.